

E8 Proving Trigonometric identities A level only

Name: _____

Class: _____

Date: _____

Time: **118 minutes**

Marks: **98 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) Prove the identity $\frac{\sin 2x}{1 + \tan^2 x} \equiv 2 \sin x \cos^3 x$ (3)

- (b) Hence find $\int \frac{4 \sin 4\theta}{1 + \tan^2 2\theta} d\theta$ (6)
- (Total 9 marks)

Q2.

- Prove the identity $\cot^2 \theta - \cos^2 \theta \equiv \cot^2 \theta \cos^2 \theta$ (6)
- (Total 3 marks)

Q3.

- (a) Show that $\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)}$ can be written as $\operatorname{cosec}^2 x$. (3)

- (b) Hence solve the equation $\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \operatorname{cosec} x + 3$ giving the values of x to the nearest degree in the interval $-180^\circ < x < 180^\circ$. (6)

- (c) Hence solve the equation $\frac{\sec^2(2\theta - 60^\circ)}{(\sec(2\theta - 60^\circ) + 1)(\sec(2\theta - 60^\circ) - 1)} = \operatorname{cosec}(2\theta - 60^\circ) + 3$ giving the values of θ to the nearest degree in the interval $0^\circ < \theta < 90^\circ$. (2)
- (Total 11 marks)

Q4.

Prove that, for all values of x , the value of the expression

$$(3 \sin x + \cos x)^2 + (\sin x - 3 \cos x)^2$$

is an integer and state its value.

(Total 4 marks)

Q5.

- (a) (i) Express $6 \sin \theta + 8 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give your value for α to the nearest 0.1° .

(2)

- (ii) Hence solve the equation $6 \sin 2x + 8 \cos 2x = 7$, giving all solutions to the nearest 0.1° in the interval $0^\circ < x < 360^\circ$.

(4)

- (b) (i) Prove the identity $\frac{\sin 2x}{1 - \cos 2x} = \frac{1}{\tan x}$.

(4)

- (ii) Hence solve the equation

$$\frac{\sin 2x}{1 - \cos 2x} = \tan x$$

giving all solutions in the interval $0^\circ < x < 360^\circ$.

(4)

(Total 14 marks)

Q6.

- (a) By writing $\sin 3x$ as $\sin(x + 2x)$, show that $\sin 3x = 3 \sin x - 4 \sin^3 x$ for all values of x .

(5)

- (b) Hence, or otherwise, find $\int \sin^3 x \, dx$.

(3)

(Total 8 marks)

Q7.

- (a) Express $\cos 2x$ in terms of $\sin x$.

(1)

- (b) (i) Hence show that $3 \sin x - \cos 2x = 2 \sin^2 x + 3 \sin x - 1$ for all values of x .

(2)

- (ii) Solve the equation $3 \sin x - \cos 2x = 1$ for $0^\circ < x < 360^\circ$.

(4)

- (c) Use your answer from part (a) to find $\int \sin^2 x \, dx$.

(2)

(Total 9 marks)

Q8.

- (a) Use the identity

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

to express $\tan 2x$ in terms of $\tan x$.

(2)

- (b) Show that

$$2 - 2 \tan x - \frac{2 \tan x}{\tan 2x} = (1 - \tan x)^2$$

for all values of x , $\tan 2x \neq 0$.

(4)

(Total 6 marks)

Q9.

- (a) Write down $\int \sec^2 x \, dx$.

(1)

- (b) Given that $y = \frac{\cos x}{\sin x}$, use the quotient rule to show that $\frac{dy}{dx} = -\operatorname{cosec}^2 x$.

(4)

- (c) Prove the identity $(\tan x + \cot x)^2 = \sec^2 x + \operatorname{cosec}^2 x$.

(3)

- (d) Hence find $\int_{0.5}^1 (\tan x + \cot x)^2 \, dx$ giving your answer to two significant figures.

(4)

(Total 12 marks)

Q10.

- (a) Describe the geometrical transformation that maps the curve with equation $y = \sin x$ onto the curve with equation:

(i) $y = 2 \sin x$;

(2)

- (ii) $y = -\sin x$; (2)
- (iii) $y = \sin(x - 30^\circ)$. (2)
- (b) Solve the equation $\sin(\theta - 30^\circ) = 0.7$, giving your answers to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$. (3)
- (c) Prove that $(\cos x + \sin x)^2 + (\cos x - \sin x)^2 = 2$. (4)
- (Total 13 marks)

Q11.

- (a) (i) Express $\sin 2x$ in terms of $\sin x$ and $\cos x$. (1)
- (ii) Express $\cos 2x$ in terms of $\cos x$. (1)
- (b) Show that
 $\sin 2x - \tan x = \tan x \cos 2x$
for all values of x . (3)
- (c) Solve the equation $\sin 2x - \tan x = 0$, giving all solutions in degrees in the interval $0^\circ < x < 360^\circ$. (4)
- (Total 9 marks)